

Course : Data Analytics & Data Science
Assignment : Research Assignment 1.

SECTION A : Data Fundamentals.

1. A database is where data is stored, this data is organised, it can be searched, updated and managed.

- Example : a) An e-commerce site like Takealot uses database to store customers details, products, orders, wishlists, inventory and history of payments. When someone places the order the website reads and writes several pieces of data at once. A database keeps all of the processes reliable.
- b) Parts department uses a CRM system that stores inventory of parts, alerts for stock refill when parts are depleted.

2. Data Warehouse → A centralised system that stores cleaned, historical organised data pulled from multiple sources. It is then made effective for querying, analysis and reporting.

Database and data warehouse are both storage systems but they serve a different purposes and are effective for different type of operations. Database is designed for Online Transaction Processing, handling large volumes of simple realtime transactions, stored data from a single source such as customer orders. They can be relational or non relational depending on the structure and query needs.

Data Warehouse is built on Online Analytical Processing enabling complex queries and analysis across multiple data sources. Data warehouse are effective for read-heavy workloads and support advanced analytics, reporting, and business Intelligence.